

Kinks, Breathers, and the Rest-Kink Spectrum in the Sine–Gordon Equation

HCS Research Program

29 August 2026 (revision 0)

Abstract

We give a source-local theorem for the one-dimensional sine–Gordon equation $u_{tt} - u_{xx} + \sin u = 0$. The declared coherent families close exactly: finite-energy monotone travelling heteroclinics in every integer vacuum layer are the subluminal Lorentz kink and antikink, while the zero-charge breathers have an explicit rest frequency and a Lorentz energy–momentum ledger. At the rest kink the Hessian is the factored Pöschl–Teller operator with translation kernel and essential edge one. The period $2\pi/\Omega$ is a comoving breather clock, not a fixed-laboratory-point period after a nonzero boost. This is not a classification of every finite-energy solution and it is not an arithmetic or Hilbert–Pólya construction.

1 Frozen model and travelling heteroclinics

We use the potential $V(u) = 1 - \cos u$, with vacua $u = 2\pi k$, and define

$$E[u] = \int_{\mathbb{R}} \left[\frac{1}{2}(u_t^2 + u_x^2) + 1 - \cos u \right] dx, \quad P[u] = - \int_{\mathbb{R}} u_t u_x dx. \quad (1)$$

The minus sign in P is fixed here and in the certificate. We begin with the unrestricted ansatz $u(x, t) = U(x - vt)$, whose profile equation is $(v^2 - 1)U'' + \sin U = 0$; the speed inequality in the theorem justifies the Lorentz coordinate $\xi = \gamma_v(x - vt - x_0)$.

Theorem 1 (kink closure). *Every nonconstant finite-energy monotone travelling heteroclinic in the declared family has an integer vacuum layer $k \in \mathbb{Z}$ and, up to translation, is*

$$U_{k,+}(\xi) = 2\pi k + 4 \arctan(e^\xi), \quad U_{k,-}(\xi) = 2\pi k + 4 \arctan(e^{-\xi}), \quad (2)$$

and necessarily $|v| < 1$. Their charges are $Q = +1$ and $Q = -1$, and

$$E = 8\gamma_v, \quad P = 8\gamma_v v, \quad E^2 - P^2 = 64. \quad (3)$$

Proof. Multiplying the unrestricted profile equation $(v^2 - 1)U'' + \sin U = 0$ by U' and using the vacuum limits gives

$$\frac{1}{2}(v^2 - 1)(U')^2 = \cos U - 1.$$

The right side is nonpositive and a nonconstant heteroclinic has $U' \neq 0$ on an open set, hence $v^2 - 1 < 0$ and $|v| < 1$. In the resulting Lorentz coordinate, the first integral is $\frac{1}{2}(U')^2 = 1 - \cos U$. Monotonicity and the 2π -periodicity of the potential select the integer-layer profiles (2); the opposite sign is the antikink. In the canonical $k = 0$ rest frame $U'_+ = 2 \operatorname{sech} \xi$ and $\int 4 \operatorname{sech}^2 \xi d\xi = 8$. Lorentz change of variables gives (3) with the convention (1). $\square$

2 Baseline breather identity

The second declared coherent family already appears at the baseline stage. If $0 < \Omega < 1$ and $\eta^2 + \Omega^2 = 1$, then

$$u_B(x, t) = 4 \arctan \left(\frac{\eta \sin(\Omega t)}{\Omega \cosh(\eta x)} \right) \quad (4)$$

solves the equation by the rational identity $\sin(4 \arctan q) = 4q(1 - q^2)/(1 + q^2)^2$. Its spatial limits agree, so $Q = 0$, while the rest-frame clock is $2\pi/\Omega$. This baseline records the exact formulas only; boosts, spectral details, and the reproducibility receipt are deliberately added in the later revision rounds.

The two families therefore have different geometric roles from the outset: the kink is a heteroclinic carrying charge, whereas the breather returns in a continuous frequency family. A continuous frequency label is not an isolated primitive-periodic owner.

Source note

The classical fluxon context is recorded by McLaughlin and Scott [1]; all displayed formulas and limits are re-derived in this source-local artifact, and the citation is not a priority or completeness claim.

References

- [1] J. M. McLaughlin and A. C. Scott, “Perturbation analysis of fluxon dynamics,” *Physical Review A* 18(4), 1652–1680 (1978). DOI: 10.1103/PhysRevA.18.1652.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, primitive-orbit, or Hilbert–Pólya claim. **Data and code.** The exact profiles, ledgers, spectrum, independent checks, and build instructions are released with HCS-C236. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked displayed claims and metadata. This is not external peer review.